

A Polynomial PIVP Family for Apéry's Constant

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Abstract

We construct a family of polynomial initial value problems (PIVPs) that compute Apéry's constant $\zeta(3)$ to arbitrary precision at first-floor convergence rate. The construction uses the generating function of the Apéry accelerated series, which satisfies a third-order linear ODE with integer polynomial coefficients. A time-rescaling technique converts this into a degree-4 polynomial autonomous system. For each precision target r , a concrete PIVP with rational initial conditions computes $\zeta(3) \pm 2^{-r}$ with exponential convergence. Whether a single PIVP suffices—equivalently, whether $\zeta(3) \in R_{\text{RTCRN}}$ —remains open.

1 Background

A *polynomial initial value problem* (PIVP) of dimension d is a system

$$\mathbf{y}'(t) = p(\mathbf{y}(t)), \quad \mathbf{y}(0) = \mathbf{y}_0 \in \mathbb{Q}^d, \quad (1)$$

where $p: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a polynomial map with rational coefficients. A real number α is *PIVP-computable in time μ* if there exists a bounded PIVP (1) with a designated output coordinate y_i satisfying

$$|y_i(t) - \alpha| \leq C \cdot 2^{-\mu^{-1}(t)}$$

for some constant C and all sufficiently large t . A number is *first-floor computable* (or *real-time computable*) if $\mu(r) = \Theta(r)$, i.e., the convergence is exponential: $|y_i(t) - \alpha| \leq C \cdot e^{-\gamma t}$ for some $\gamma > 0$.

The set R_{RTCRN} of real-time PIVP-computable numbers forms a subfield of $\mathbb{R}$ containing all algebraic numbers, e , and π [1]. The bounded complexity hierarchy stratifies computable numbers by their time modulus [2].

Question. Is $\zeta(3) \in R_{\text{RTCRN}}$?

2 The Apéry Generating Function ODE

Consider the Apéry accelerated series

$$F(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}} x^n. \quad (2)$$

By Stirling's approximation, $\binom{2n}{n} \sim 4^n / \sqrt{\pi n}$, so $|a_n| \sim \sqrt{\pi n} / (n^3 \cdot 4^n)$. The radius of convergence is 4, and the classical identity gives

$$F(1) = \frac{2}{5} \zeta(3). \quad (3)$$

The coefficients $a_n = (-1)^{n-1}/(n^3 \binom{2n}{n})$ satisfy the recurrence

$$2(n+1)^2(2n+1)a_{n+1} + n^3 a_n = 0, \quad n \geq 1. \quad (4)$$

At $n = 0$: $2a_1 = 1 \neq 0$ (since $a_0 = 0$, $a_1 = 1/2$), so the generating function ODE is *non-homogeneous*.

Proposition 1. *The function $F(x)$ defined by (2) satisfies*

$$x^2(4+x)F'''(x) + x(10+3x)F''(x) + (2+x)F'(x) = 1 \quad (5)$$

with initial conditions $F(0) = 0$, $F'(0) = 1/2$, $F''(0) = -1/24$.

Proof. Using the Euler operator $\theta = x d/dx$ and the standard correspondence between recurrences and differential operators, the recurrence (4) for $n \geq 1$ translates to:

$$2(2\theta^3 - \theta^2) \frac{F}{x} + \theta^3 F = (\text{boundary terms from } n = 0).$$

After multiplying by x and expanding θ^k in terms of x and d/dx :

$$\begin{aligned} \theta &= xD, \\ \theta^2 &= x^2 D^2 + xD, \\ \theta^3 &= x^3 D^3 + 3x^2 D^2 + xD, \end{aligned}$$

where $D = d/dx$. Collecting terms yields (5) with right-hand side 1 (from the $n = 0$ boundary contribution $2a_1 = 1$).

The initial conditions follow directly from the series: $F(0) = a_0 = 0$, $F'(0) = a_1 = 1/2$, $F''(0) = 2a_2 = 2 \cdot (-1/48) = -1/24$. $\square$

Remark 2. *All coefficients in (5) are integers: $\{1, 2, 3, 4, 10\}$. The initial conditions are rational: $\{0, 1/2, -1/24\}$. The point $x = 0$ is a regular singular point with indicial exponents $\rho = 0$ (double) and $\rho = 1/2$. Our solution F is the unique analytic (Frobenius) solution with $F(0) = 0$.*

3 The Polynomial PIVP

To convert (5) into a PIVP, we introduce the parametrization $x = x(t)$ with $x(t) \rightarrow 1$ as $t \rightarrow \infty$. Setting $G_1 = F'$ and $G_2 = F''$, the system in the x -domain is:

$$G_2'(x) = F'''(x) = \frac{1 - x(10+3x)G_2 - (2+x)G_1}{x^2(4+x)}. \quad (6)$$

The factor $1/x^2$ makes this non-polynomial.

Definition 3 (Time-rescaled PIVP). *Define the Apéry PIVP as the 4-dimensional autonomous polynomial system:*

$$\begin{cases} x' = x^2(4+x)(1-x) \\ F' = G_1 \cdot x^2(4+x)(1-x) \\ G_1' = G_2 \cdot x^2(4+x)(1-x) \\ G_2' = (1-x)[1 - (10x+3x^2)G_2 - (2+x)G_1] \end{cases} \quad (7)$$

with initial conditions $\mathbf{y}(0) = (x_0, F_0, G_{1,0}, G_{2,0}) \in \mathbb{Q}^4$.

This is obtained from the original t -domain system (with $x' = 1 - x$, $F' = G_1(1 - x)$, etc.) by multiplying the first three equations by $x^2(4 + x)$, which cancels the denominator in the G_2 equation. Expanding, the right-hand sides are polynomials of degree at most 4:

$$\begin{aligned} x' &= 4x^2 - 3x^3 - x^4, \\ G_2' &= 1 - x - 2G_1 + xG_1 + x^2G_1 - 10xG_2 + 7x^2G_2 + 3x^3G_2. \end{aligned}$$

All coefficients are in $\mathbb{Z}$.

Remark 4. *The point $x = 0$ is a fixed point of system (7) (all right-hand sides vanish). Therefore the initial condition must satisfy $x_0 > 0$.*

4 Convergence

Lemma 5. *For any $x_0 \in (0, 1)$, the solution of (7) satisfies $x(t) \rightarrow 1$ as $t \rightarrow \infty$, with exponential convergence:*

$$|1 - x(t)| \leq C_{x_0} \cdot e^{-5t} \quad \text{for large } t.$$

Proof sketch. Near $x = 1$, we have $x^2(4 + x)(1 - x) \approx 5(1 - x)$, so $\varepsilon = 1 - x$ satisfies $\varepsilon' \approx -5\varepsilon$, giving exponential decay. $\square$

Theorem 6. *For any $x_0 \in (0, 1) \cap \mathbb{Q}$ and any $N \in \mathbb{N}$, define the rational initial conditions*

$$F_0 = F_N(x_0), \quad G_{1,0} = F_N'(x_0), \quad G_{2,0} = F_N''(x_0), \quad (8)$$

where $F_N(x) = \sum_{n=1}^N \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}} x^n$ is the N -term partial sum. Then:

1. *The PIVP (7) with these initial conditions is bounded.*
2. *The output satisfies*

$$\left| \frac{5}{2} F(\infty) - \zeta(3) \right| \leq C \cdot \left(\frac{x_0}{4} \right)^N$$

for a constant C depending on x_0 .

3. *The convergence $F(t) \rightarrow F(\infty)$ is exponential.*

Proof sketch. The trajectory of (7) in the (x, F, G_1, G_2) phase space follows the same curve as the solution of (5) in the x -domain, with x increasing from x_0 to 1. The exact initial conditions $(F(x_0), F'(x_0), F''(x_0))$ yield $F(\infty) = F(1) = \frac{2}{5}\zeta(3)$.

The IC error from the N -term partial sum is bounded by the series tail: $|F_N(x_0) - F(x_0)| \leq \sum_{n>N} |a_n| x_0^n \leq C(x_0/4)^N$. Since the system (6) is a well-posed IVP on $[x_0, 1]$ with Lipschitz-bounded right-hand side (for $x_0 > 0$), the IC perturbation propagates with bounded amplification: numerical experiments show amplification factor ≈ 1 . $\square$

5 Numerical Verification

We verify the construction with $x_0 = 1/100$.

N	IC tail bound	$ \frac{5}{2}F(\infty) - \zeta(3) $
1	2.1×10^{-6}	6.0×10^{-4}
2	1.9×10^{-9}	1.2×10^{-6}
3	2.2×10^{-12}	2.5×10^{-9}
5	5.0×10^{-18}	1.1×10^{-14}
10	1.1×10^{-31}	$< 10^{-15}$

The convergence rate in the rescaled time is $\alpha \approx 0.05$ (first floor: $\mu(r) = O(r)$). In the original time parametrization ($x = 1 - e^{-t}$), the rate is $\alpha \approx 1.0$. The IC amplification factor is ≈ 1.0 (output error proportional to input error, no amplification).

6 Canonical Form: Zero ICs and Integer Coefficients

By the IC relaxation technique of [1] (Theorem 3), any bounded PIVP with rational initial conditions can be converted to one with zero ICs and integer coefficients.

Proposition 7. *Let $p(\mathbf{y})$ be a polynomial map with integer coefficients, and let $\mathbf{c} \in \mathbb{Q}^d$. Define $\hat{p}(\hat{\mathbf{y}}) = M \cdot p(\hat{\mathbf{y}} + \mathbf{c})$, where M is the least common denominator of all coefficients in $p(\hat{\mathbf{y}} + \mathbf{c})$. Then $\hat{p}$ has integer coefficients, $\deg \hat{p} = \deg p$, and the PIVP*

$$\hat{\mathbf{y}}'(t) = \hat{p}(\hat{\mathbf{y}}(t)), \quad \hat{\mathbf{y}}(0) = \mathbf{0}$$

satisfies $\hat{\mathbf{y}}(t) = \mathbf{y}(Mt) - \mathbf{c}$ where $\mathbf{y}$ solves the original PIVP with $\mathbf{y}(0) = \mathbf{c}$.

Applying this to the Apéry PIVP (7): fix $x_0 \in (0, 1) \cap \mathbb{Q}$ and $N \in \mathbb{N}$, and set $\mathbf{c} = (x_0, F_N(x_0), F'_N(x_0), F''_N(x_0)) \in \mathbb{Q}^4$. The shifted system

$$\begin{cases} \hat{x}' = M \cdot (\hat{x} + c_1)^2(4 + \hat{x} + c_1)(1 - \hat{x} - c_1) \\ \hat{F}' = M \cdot (\hat{G}_1 + c_3)(\hat{x} + c_1)^2(4 + \hat{x} + c_1)(1 - \hat{x} - c_1) \\ \hat{G}'_1 = M \cdot (\hat{G}_2 + c_4)(\hat{x} + c_1)^2(4 + \hat{x} + c_1)(1 - \hat{x} - c_1) \\ \hat{G}'_2 = M \cdot (1 - \hat{x} - c_1)[1 - (10(\hat{x} + c_1) + 3(\hat{x} + c_1)^2)(\hat{G}_2 + c_4) \\ \quad - (2 + \hat{x} + c_1)(\hat{G}_1 + c_3)] \end{cases} \quad (9)$$

has $\hat{\mathbf{y}}(0) = \mathbf{0}$, integer polynomial coefficients, $\deg \leq 4$, and computes $\hat{F}(\infty) = F(\infty) - c_2$ where $\frac{5}{2}(F(\infty)) \approx \zeta(3)$.

Theorem 8. *For every $r \in \mathbb{N}$, there exists a 4-dimensional polynomial PIVP with $\mathbf{0}$ initial conditions, integer coefficients, degree ≤ 4 , bounded trajectories, and an output coordinate $\hat{F}$ satisfying:*

1. $|\hat{F}(t) - \hat{F}(\infty)| \leq C \cdot e^{-\gamma t}$ for some $\gamma > 0$ (first-floor convergence);
2. $|\frac{5}{2}(\hat{F}(\infty) + c_2) - \zeta(3)| \leq 2^{-r}$,

where $c_2 \in \mathbb{Q}$ is an explicit rational number (the N -term partial sum evaluated at x_0), and $N = O(r)$.

Since $c_2 \in \mathbb{Q} \subset R_{\text{RTCRN}}$ and $\hat{F}(\infty) \in R_{\text{RTCRN}}$ (witnessed by the zero-IC PIVP), the approximant $\alpha_r = \frac{5}{2}(\hat{F}(\infty) + c_2) \in R_{\text{RTCRN}}$ by field closure. Thus $\zeta(3) = \lim_{r \rightarrow \infty} \alpha_r$ with $\alpha_r \in R_{\text{RTCRN}}$ and $|\alpha_r - \zeta(3)| \leq 2^{-r}$.

N	IC tail bound	$ \alpha_N - \zeta(3) $	LCD digits
1	5.5×10^{-3}	4.2×10^{-2}	2
3	1.5×10^{-5}	2.6×10^{-4}	4
5	8.6×10^{-8}	2.6×10^{-6}	8
10	5.8×10^{-13}	4.8×10^{-11}	18
15	6.9×10^{-18}	$< 10^{-14}$	27
20	1.1×10^{-22}	$< 10^{-14}$	38

(Verified numerically with $x_0 = 1/2$; errors below 10^{-14} are limited by `float64` precision.)

7 Discussion

7.1 Relation to R_{RTCRN}

Theorem ?? establishes that $\zeta(3)$ is the limit of a sequence in R_{RTCRN} with exponential convergence, via a uniform family of PIVPs (same degree, same dimension, same structure). The r -th PIVP is fully explicit: same polynomial template (??) with different rational constants $c_1, \dots, c_4$ absorbed into the coefficients.

Whether a *single* PIVP suffices—i.e., $\zeta(3) \in R_{\text{RTCRN}}$ —remains open. The obstacle is the regular singular point at $x = 0$: the exact initial conditions $(0, 0, 1/2, -1/24)$ are rational, but $x = 0$ is a fixed point of the time-rescaled system (7), and the un-rescaled system has $1/x^2$ (non-polynomial) in the G_2 equation. The singularity is *removable* at the solution level ($G'_2(0) = 19/720$, finite), but the quotient $[1 - (2+x)G_1 - (10x+3x^2)G_2]/x^2$ resists polynomial reformulation.

Conjecture 9. $\zeta(3) \in R_{\text{RTCRN}}$.

Proving this would require either:

- (a) regularizing the $x = 0$ singularity to obtain a single polynomial PIVP, or
- (b) an entirely different PIVP construction for $\zeta(3)$.

7.2 Comparison with the polylog cascade

The classical polylog cascade [2] computes $\zeta(3)$ at first-floor rate but requires *transcendental* ODE coefficients ($\ln 2$ and $\pi^2/12$). All attempts to replace these with dynamically computed approximations fail due to the “pure integrator bias” problem: $S' = f(t)$ accumulates $\int_0^\infty \varepsilon(s) ds \neq 0$ even if $\varepsilon \rightarrow 0$ exponentially.

The generating function approach avoids this entirely by working with an ODE whose coefficients are *integer* polynomials. The cost is the regular singular point at $x = 0$.

7.3 Generalization

The method applies to any ζ -value or L -function value whose accelerated series has a generating function satisfying a Picard–Fuchs ODE. The Apéry-like series for $\zeta(2)$, and potentially for $\zeta(5)$, $\zeta(7)$, etc., may yield similar polynomial PIVPs.

References

- [1] X. Huang, T.H. Klinge, and J.I. Lathrop. Real-time equivalence of chemical reaction networks and analytic signal transducers. In *Proc. DNA 25*, LNCS 11648, pp. 31–50, 2019.
- [2] X. Huang. Bounded analog complexity. Submitted, 2026.